

A CONSTRUCTION OF THE EXTREMAL TORUS FIXED POINT, AND A CRITERION FOR SIMPLE SPECTRUM

ABSTRACT. Let Γ be a simply-laced Dynkin diagram, $\lambda = \omega_\kappa$ a fundamental weight and w an element of the Weyl group. When $\mu = w\lambda$ the Nakajima quiver variety $\mathcal{M}(v, w)$ with $\lambda - \mu = \sum_i v_i \alpha_i$ is a single point, carrying a unique T -fixed representation $V = \bigoplus_i V_i$. Dinkins, Karpov and Krylov raise, in §5.4.2 of [4], the problem of characterising combinatorially those w for which T acts on every V_i with simple spectrum, in which case the fixed point is encoded by a coloured poset and the vertex function is a sum over reverse plane partitions.

We solve this by producing V itself. Let Π be the preprojective algebra of Γ and I_κ the injective envelope of the simple S_κ . Reading a reduced word for w from the right and recording at each step the integer $c = \langle \alpha_i^\vee, \mu_{\text{cur}} \rangle$ produces a sequence σ of vertices with multiplicities. We prove that

$$V \cong \text{soc}_\sigma(I_\kappa),$$

the iterated socle of Geiss–Leclerc–Schröer taken along σ . The proof is uniform in type and rests on an identity, derived from Crawley-Boevey’s formula for Ext^1 over a preprojective algebra, together with a squeeze against $\dim I_\kappa = \lambda - w_0 \lambda$. Simple spectrum then becomes the assertion that no two elements of the resulting coloured graded set share a colour and a degree, which is decidable in polynomial time from any reduced word. In type D_n , for non-spin λ , we make the answer completely explicit: writing $\mu = \sum_{i \in S} \eta_i e_i$ in Bourbaki coordinates, the spectrum is simple if and only if $r + 2m \leq 2$, where m counts the minus signs among the first $n - 2$ coordinates and $r = |S \cap \{n - 1, n\}|$. The proof of this last statement runs through a strand decomposition of I_κ under the diagram automorphism exchanging the two leaves, which reduces it to an explicit inequality between two integer sequences. We also record an element of D_4 that is fully commutative but does not have simple spectrum, showing that the sufficient condition considered in [4, Ex. 5.12] fails in both directions.

1. INTRODUCTION

1.1. The problem. Fix a simply-laced Dynkin diagram Γ with vertex set I , Cartan matrix $C = (a_{ij})$, simple roots α_i , fundamental weights ω_i and Weyl group W . Let $\mathcal{M}(v, w)$ denote the Nakajima quiver variety attached to Γ with dimension vector v and framing w , and write $\lambda = \sum_i w_i \omega_i$, $\mu = \lambda - \sum_i v_i \alpha_i$. When μ is an extremal weight of the irreducible representation of highest weight λ — that is, $\mu = w\lambda$ for some $w \in W$ — the variety $\mathcal{M}(v, w)$ is a single point [7, 8], and that point is a representation $V = \bigoplus_{i \in I} V_i$ of the doubled quiver satisfying the moment map equations, unique up to isomorphism.

The point is fixed by the torus T acting on the quiver data, so each V_i carries a T -weight decomposition. Dinkins, Karpov and Krylov show [4, Thm. 5.6, Thm. 5.7] that when T acts on every V_i with *simple spectrum* — all weights distinct — the fixed point is encoded by a finite coloured poset, and the associated vertex function is a generating function over reverse plane partitions on that poset. They then write, in §5.4.2:

“We consider it an interesting open problem to find a combinatorial characterization of such w .”

They rule out three natural guesses. Dominant minuscule elements are too restrictive [4, Ex. 5.10]; so are minuscule elements [4, Ex. 5.11]; and so are fully commutative elements [4, Ex. 5.12]. Each example exhibits a w with simple spectrum lying outside the class.

1.2. Results. Our main theorem answers the question by constructing V explicitly, uniformly in the type, from any reduced word for w .

Let $\Pi = \Pi(\Gamma)$ be the preprojective algebra over a field, S_i the simple module at i , and I_i the injective envelope of S_i . For a Π -module M and $j \in I$ let $\text{soc}_{(j)}(M)$ be the sum of all submodules of M isomorphic to S_j , and for a sequence $\sigma = (j_1, \dots, j_p)$ let $\text{soc}_\sigma(M)$ be the iterated socle of Geiss–Leclerc–Schröer [6, §2.4], defined by $M_0 = 0$ and $M_q/M_{q-1} = \text{soc}_{(j_q)}(M/M_{q-1})$.

Definition 1.1 (the c -trace). Let $w = s_{i_1} \cdots s_{i_\ell}$ be a reduced word. Set $\mu^{(\ell)} = \lambda$ and, for $p = \ell, \ell - 1, \dots, 1$,

$$c_p = \langle \alpha_{i_p}^\vee, \mu^{(p)} \rangle, \quad \mu^{(p-1)} = \mu^{(p)} - c_p \alpha_{i_p} = s_{i_p} \mu^{(p)}.$$

The c -weighted sequence of the word is $\sigma = (i_\ell^{c_\ell}, i_{\ell-1}^{c_{\ell-1}}, \dots, i_1^{c_1})$, each letter repeated according to its multiplicity, in this order.

Note $\mu^{(0)} = w\lambda = \mu$ and $\sum_{p: i_p=i} c_p = v_i$, so the c -trace recomputes the dimension vector. The invariant $c_{\max} = \max_p c_p$ equals 1 exactly when w is λ -minuscule in the sense of [4, Def. 5.1].

Theorem 1.2 (Main Theorem). *Let Γ be simply-laced Dynkin, $\lambda = \omega_\kappa$, $w \in W$ of minimal length in wW_λ , and σ the c -weighted sequence of any reduced word for w . Then*

$$\underline{\dim} \text{soc}_\sigma(I_\kappa) = v = \lambda - w\lambda,$$

and consequently

$$V \cong \text{soc}_\sigma(I_\kappa)$$

as graded representations of the doubled quiver. In particular the isomorphism class of $\text{soc}_\sigma(I_\kappa)$ does not depend on the chosen reduced word.

The grading on the right is the path-length grading of I_κ ; Proposition 2.6 identifies it with the T -weight grading, and §5 explains how to read off the coloured poset of [4, §5.4] from it. The criterion is then immediate.

Corollary 1.3. *T acts on every V_i with simple spectrum if and only if the coloured graded set underlying $\text{soc}_\sigma(I_\kappa)$ has no repeated (colour, degree) pair, i.e. $\dim \text{soc}_\sigma(I_\kappa)_i[d] \leq 1$ for all $i \in I$ and all d .*

The construction is effective: I_κ has dimension $\lambda - w_0\lambda$, independent of w , and the socle steps are kernel computations of size bounded by the graded pieces of I_κ , so the criterion is decidable in time polynomial in $|I|$ and $\ell(w)$ from any reduced word.

Two refinements make the answer sharper. First, only one vertex needs to be inspected.

Theorem 1.4. *Let t be the trivalent node of Γ (types D and E). If $\dim V_t[d] \leq 1$ for all d , then $\dim V_i[d] \leq 1$ for all i and d .*

Theorem 1.4 follows in both types from a single inward-injectivity lemma about I_κ (Lemma 7.1, §7). In type E that lemma is a *complete* finite computation, since E_6 , E_7 and E_8 have only finitely many triples (fundamental weight, vertex, degree); in type D it is proved uniformly in the rank (Lemma 7.2). Remark 6.4 records why the more obvious argument, containment of multiplicity-2 windows, fails in both types. In types D and E this reduces the criterion to a single vertex; in type A there is no trivalent node, but there the question is settled outright:

Proposition 1.5. *If λ is minuscule then, for w of minimal length in wW_λ , $c_p = 1$ for every step of every reduced word, w is λ -minuscule, and T acts with simple spectrum. In particular the spectrum is always simple in type A , for the spin weights ω_{n-1}, ω_n of D_n , for ω_1 and ω_6 of E_6 , and for ω_7 of E_7 .*

Second, in type D the criterion becomes a statement about the signs of μ with no reference to reduced words, heaps, or fully commutative elements.

Theorem 1.6 (type D). *Let $\Gamma = D_n$, $t = n - 2$, and $\lambda = \omega_k$ with $k \leq n - 2$. Write $\mu = w\lambda = \sum_{i \in S} \eta_i e_i$ in Bourbaki coordinates, $|S| = k$, $\eta_i = \pm 1$. Put*

$$m = \#\{i \leq n - 2 : i \in S, \eta_i = -1\}, \quad r = |S \cap \{n - 1, n\}|.$$

Then T acts with simple spectrum if and only if $r + 2m \leq 2$; equivalently, iff $m = 0$, or $m = 1$ and $r = 0$. For the spin weights the spectrum is always simple. The proof runs through Proposition 6.5, which Theorem 6.18 establishes; see Remark 6.6 for the route.

Finally we record a counterexample in the direction opposite to [4, Ex. 5.12].

Example 1.7. In D_4 take $w = s_2 s_4 s_3 s_1 s_2$, so $v = (1, 3, 1, 1)$. This w is fully commutative, but the spectrum is *not* simple: the path-length graded dimensions of V inside I_2 are $\{2 : 1\}$ in degree 4, $\{1 : 1, 3 : 1, 4 : 1\}$ in degree 3 and $\{2 : 2\}$ in degree 2, so, read from the top down as in the diagrams of [4], the rows are [2], [1, 3, 4], [2, 2]. Together with [4, Ex. 5.12], which exhibits a w with simple spectrum that is not fully commutative, this shows that full commutativity is neither necessary nor sufficient.

1.3. Method. The proof of Theorem 1.2 is short and does not proceed case by case. It has two ingredients. The first is an exact formula for the size of a single socle step, valid for an arbitrary submodule of I_κ :

Lemma 1.8 (§3). *Let $X \subseteq I_\kappa$ be a submodule, $v' = \underline{\dim} X$, $\mu' = \lambda - \sum_i v'_i \alpha_i$. Then for every $i \in I$*

$$\dim \text{soc}_{(i)}(I_\kappa/X) = \langle \alpha_i^\vee, \mu' \rangle + \dim \text{top}_i(X),$$

where $\text{top}_i(X) = \text{Hom}_\Pi(X, S_i)$.

This comes from applying $\text{Hom}(S_i, -)$ to $0 \rightarrow X \rightarrow I_\kappa \rightarrow Q \rightarrow 0$, using $\text{Ext}^1(S_i, I_\kappa) = 0$, and then Crawley-Boevey's formula $\dim \text{Ext}_\Pi^1(M, N) = \dim \text{Hom}(M, N) + \dim \text{Hom}(N, M) - (\underline{\dim} M, \underline{\dim} N)$ [3].

The second ingredient is a squeeze. Since $\dim \text{top}_i(X) \geq 0$, each socle step grows X by at least c ; summing the c 's along a reduced word for the longest element gives exactly $\lambda - w_0 \lambda$, which is precisely $\underline{\dim} I_\kappa$. As $\text{soc}_\sigma(I_\kappa) \subseteq I_\kappa$ the two bounds meet, so *every* step contributes exactly c and every top-term vanishes. Restricting to the prefix corresponding to w gives Theorem 1.2, once uniqueness of the submodule with w -extremal dimension vector [8, Prop. 4.9] identifies the result with V .

1.4. Verification. Every statement in this paper is accompanied by machine verification against an independent solver that enumerates graded submodules directly. In particular the construction reproduces the three coloured posets printed in [4, pp. 82–83], Hasse edges included, and the criterion agrees with the solver on 2179 elements across types A , D and E — every fundamental weight of A_3 – A_6 , D_4 – D_7 , E_6 , together with $E_7 \omega_1$ and $E_7 \omega_7$, whose Weyl orbits have at most 400 elements. See §8.

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2. SETUP

2.1. The preprojective algebra. Let Q be a quiver with underlying graph Γ and let $\overline{Q}$ be its double, with a^* the reverse of a . The preprojective algebra is

$$\Pi = \Pi(\Gamma) = \mathbb{k}\overline{Q} / \left(\sum_a (aa^* - a^*a) \right),$$

where $\mathbb{k}$ is any field. Every argument below is characteristic-free; Crawley-Boevey's formula, which we use in §3, holds over an arbitrary field. The computer verifications of §8 were run over $\mathbb{F}_p$ for $p = 5, 7, 11, 13$ with identical results. A Π -module is the same thing as a representation of $\overline{Q}$ satisfying the moment map equation at every vertex, which is exactly the datum carried by a point of $\mathcal{M}(v, w)$ with vanishing stability-free part. We use throughout the following standard facts, valid because Γ is Dynkin.

Proposition 2.1. $\Pi(\Gamma)$ is finite dimensional and self-injective. Its Nakayama permutation ν is determined by $\text{soc}(\Pi e_j) \cong S_{\nu(j)}$, and

$$I_\kappa \cong \Pi e_{\nu^{-1}(\kappa)}.$$

Moreover ν is the identity except in types A_n , D_{odd} and E_6 , where it is induced by the nontrivial diagram automorphism.

Proposition 2.2. $\dim I_\kappa = \omega_\kappa - w_0 \omega_\kappa$ in root coordinates.

Proposition 2.1 is classical; see for instance [3, §1]. Proposition 2.2 follows from the description of Πe_j as $\bigoplus_{d \geq 0} \tau^{-d} P_j$ over the path algebra of an orientation of Γ , whose total dimension vector is $\omega_j + \omega_{j^*} = \omega_j - w_0 \omega_j$; it is verified directly for every fundamental weight of $A_3, A_4, A_5, D_4, D_5, D_6, D_7, E_6$ and E_7 in §8.

Every Πe_j is graded by path length, and *this is the grading we use throughout*; it coincides with the socle-layer grading read in reverse. We write $M_i[d]$ for the degree- d part of M at vertex i , and $d_{\max}$ for the top degree of I_κ .

2.2. Iterated socles. For $j \in I$, $\text{soc}_{(j)}(M)$ is the largest submodule of M that is a direct sum of copies of S_j ; concretely it is the set of $x \in M_j$ annihilated by every arrow out of j . Given a sequence $\sigma = (j_1, \dots, j_p)$ define $\text{soc}_\sigma(M) = M_p$ where $M_0 = 0$ and M_q is the preimage in M of $\text{soc}_{(j_q)}(M/M_{q-1})$. We use only that soc_σ is a submodule and that $\text{soc}_\sigma(M) \subseteq \text{soc}_{\sigma'}(M)$ when σ is a prefix of σ' .

2.3. Extremal fixed points. We use Savage–Tingley's identification of the quiver variety with a quiver Grassmannian.

Proposition 2.3 ([8, Thm. 4.4, Prop. 4.9]; [1, 2]). $\mathfrak{L}(v, w) \cong \text{Gr}_v(\bigoplus_i I_i^{w_i})$, and if $v = \lambda - w\lambda$ is w -extremal then there is exactly one submodule of $\bigoplus_i I_i^{w_i}$ with dimension vector v .

The uniqueness has a sharper and more useful form. Baumann and Kamnitzer attach to a Π -module T a pseudo-Weyl polytope $\text{Pol}(T)$, and show [1] that it is the convex hull of the dimension vectors of the submodules of T ; Baumann, Kamnitzer and Tingley prove the *rigidity* property [2, §3.1]: *if x is a vertex of $\text{Pol}(T)$ then T has a unique subobject X with $\dim X = x$* . For $T = I_\kappa$ the polytope is the weight polytope of ω_κ reflected in the origin, so its vertices are exactly the w -extremal dimension vectors, and Proposition 2.3 is the vertex case. Two consequences we use below:

Corollary 2.4. Let $U \subseteq I_\kappa$ be a submodule. Then $\mu = \lambda - \dim U$ is a weight of the irreducible representation $L(\omega_\kappa)$, and

$$\delta(\dim U) = 2(\dim U)_\kappa - (\dim U)^\top C(\dim U) = (\lambda, \lambda) - (\mu, \mu) \geq 0,$$

with equality if and only if $\underline{\dim}U$ is a vertex of $\text{Pol}(I_\kappa)$, i.e. w -extremal. If I_κ carries two distinct submodules with the same dimension vector v , then v is not a vertex and $\delta(v) > 0$.

In type D_n with $\lambda = \omega_k$, $k \leq n - 2$, the weights of $L(\omega_k)$ are the μ with entries in $\{0, \pm 1\}$ and $|\text{supp } \mu| \leq k$ congruent to k modulo 2, so Corollary 2.4 gives $\delta(\underline{\dim}U) = k - |\text{supp } \mu|$ — verified on all 2014 graded submodules of I_κ for $D_5\omega_2$, $D_5\omega_3$, $D_6\omega_3$, $D_6\omega_4$, $D_7\omega_3$.

For $\lambda = \omega_\kappa$ the framing is $\bigoplus_i I_i^{w_i} = I_\kappa$, so Proposition 2.3 says: I_κ has a unique submodule of dimension vector v , and the fixed point V is that submodule (we use [8, Thm. 4.4] in the convention in which a point of the quiver Grassmannian is a submodule of the injective, not a quotient of it). Hence to identify V it suffices to exhibit any submodule of I_κ of the right dimension vector. That is what Theorem 1.2 does.

It remains to match the two gradings.

Lemma 2.5 (the bidegree is determined). Π is bigraded by the numbers of a - and a^* -arrows separately, the relations $\sum_a (aa^* - a^*a)$ being bihomogeneous of bidegree $(1, 1)$. For Γ a Dynkin diagram — hence a tree — the piece $(\Pi e_j)_i[d]$ is pure of a single bidegree, namely

$$p = \frac{1}{2}(d + f(i) - f(j)), \quad q = \frac{1}{2}(d - f(i) + f(j)),$$

where f is any function on the vertices with $f(\text{head}) - f(\text{tail}) = 1$ on every arrow of the chosen orientation.

Proof. Such an f exists because Γ is a tree: define it by walking out from any root; there is no cycle to obstruct consistency. An a -arrow raises f by 1 and its reverse lowers it by 1, so along any path from j to i one has $p - q = f(i) - f(j)$. With $p + q = d$ this determines p and q . $\square$

Proposition 2.6. The T -weight decomposition of V_i coincides with its path-length decomposition, up to an overall affine normalisation independent of i . In particular

$$T \text{ acts on every } V_i \text{ with simple spectrum} \iff \dim V_i[d] \leq 1 \text{ for all } i \text{ and } d.$$

Proof. Because $\lambda = \omega_\kappa$ is fundamental the framing space is one dimensional, so the framing torus acts on all of V through a single character and contributes only a global shift. The rest of T acts through the bigrading of Π : any such torus assigns to $(\Pi e_{\kappa'})_i[d]$ a weight that is a fixed integral combination $\alpha p + \beta q$ of the two bidegrees. By Lemma 2.5 both p and q are affine functions of d with i, κ' fixed, with p and q each of slope $\pm \frac{1}{2}$ in d ; hence $\alpha p + \beta q$ is an affine function of d of slope $\frac{1}{2}(\alpha + \beta)$, which is injective in d unless $\alpha + \beta = 0$. The case $\alpha + \beta = 0$ is excluded because the symplectic form has nonzero weight, that being what makes the fixed locus of T on $\mathcal{M}(v, w)$ isolated. So the weight determines and is determined by d , degree by degree.

Finally V is a graded submodule: since v is w -extremal, Proposition 2.3 gives a unique submodule of I_κ with dimension vector v , and uniqueness forces it to be stable under the whole torus, hence to carry the restricted grading. $\square$

3. THE SOCLE-STEP IDENTITY

Proof of Lemma 1.8. Write $I = I_\kappa$ and let $Q = I/X$, so that $0 \rightarrow X \rightarrow I \rightarrow Q \rightarrow 0$ is exact. Applying $\text{Hom}_\Pi(S_i, -)$ gives

$$0 \rightarrow \text{Hom}(S_i, X) \rightarrow \text{Hom}(S_i, I) \rightarrow \text{Hom}(S_i, Q) \rightarrow \text{Ext}^1(S_i, X) \rightarrow \text{Ext}^1(S_i, I).$$

The last term vanishes because I is injective. Since $\text{soc}(I_\kappa) = S_\kappa$ we have $\dim \text{Hom}(S_i, I) = \delta_{i\kappa}$, and $\dim \text{Hom}(S_i, Q) = \dim \text{soc}_{(i)}(Q)$. Therefore

$$\dim \text{soc}_{(i)}(Q) = \delta_{i\kappa} - \dim \text{soc}_{(i)}(X) + \dim \text{Ext}^1(S_i, X). \quad (1)$$

By Crawley-Boevey's formula for modules over a preprojective algebra [3, Lem. 1],

$$\dim \operatorname{Ext}_{\Pi}^1(M, N) = \dim \operatorname{Hom}(M, N) + \dim \operatorname{Hom}(N, M) - (\underline{\dim} M, \underline{\dim} N),$$

where $(\cdot, \cdot)$ is the symmetrised Tits form, so that $(\underline{\dim} M, \underline{\dim} N) = (\underline{\dim} M)^\top C (\underline{\dim} N)$. Taking $M = S_i$ and $N = X$, and noting $\operatorname{Hom}(S_i, X) = \operatorname{soc}_{(i)}(X)$ and $\operatorname{Hom}(X, S_i) = \operatorname{top}_i(X)$,

$$\dim \operatorname{Ext}_{\Pi}^1(S_i, X) = \dim \operatorname{soc}_{(i)}(X) + \dim \operatorname{top}_i(X) - (\alpha_i, v').$$

Substituting into (1) the socle terms cancel and

$$\dim \operatorname{soc}_{(i)}(Q) = \delta_{i\kappa} - (\alpha_i, v') + \dim \operatorname{top}_i(X).$$

Finally $(\alpha_i, v') = \sum_j a_{ij} v'_j = \langle \alpha_i^\vee, \sum_j v'_j \alpha_j \rangle$ and $\langle \alpha_i^\vee, \omega_\kappa \rangle = \delta_{i\kappa}$, so $\delta_{i\kappa} - (\alpha_i, v') = \langle \alpha_i^\vee, \mu' \rangle$. $\square$

Corollary 3.1. *A socle step of colour i applied to $X \subseteq I_\kappa$ enlarges X by exactly $\langle \alpha_i^\vee, \mu' \rangle + \dim \operatorname{top}_i(X)$ dimensions, all at vertex i . In particular the growth is at least $\langle \alpha_i^\vee, \mu' \rangle$, with equality if and only if S_i does not occur in the head of X .*

Remark 3.2. Lemma 1.8 also explains why repeating a colour is harmless. Suppose a step of colour i with $c = \langle \alpha_i^\vee, \mu' \rangle$ has just added exactly c dimensions, producing X' . Those c new elements lie in the head of X' , so $\dim \operatorname{top}_i(X') = c$; and the weight has become $s_i \mu'$, with $\langle \alpha_i^\vee, s_i \mu' \rangle = -c$. Lemma 1.8 gives $\dim \operatorname{soc}_{(i)}(I_\kappa/X') = -c + c = 0$: further repeats add nothing. Thus $\operatorname{soc}_\sigma$ is unchanged if each block i^c is replaced by $i^{c'}$ for any $c' \geq c$, and the c -weighted sequence may equivalently be described as “apply $\operatorname{soc}_{(i)}$ until it stabilises”.

4. PROOF OF THE MAIN THEOREM

Lemma 4.1. *Let w be of minimal length in wW_λ and $\mu = w\lambda$.*

- (i) *Every c -trace value of every reduced word for w satisfies $c_p \geq 1$.*
- (ii) *If $\mu \neq w_0\lambda$ there is a simple reflection s_i with $\langle \alpha_i^\vee, \mu \rangle \geq 1$, and for any such i one has $\ell(s_i w) = \ell(w) + 1$ with $s_i w$ again of minimal length in $s_i wW_\lambda$.*

Consequently every reduced word for w extends on the left to a reduced word for the longest minimal-length representative w_0^λ , with all c -trace values ≥ 1 .

Proof. (i) Write $u = s_{i_{p+1}} \cdots s_{i_\ell}$, so $\mu^{(p)} = u\lambda$. The word is reduced, so $\ell(s_{i_p} u) = \ell(u) + 1$ and hence $u^{-1} \alpha_{i_p} > 0$; since λ is dominant, $c_p = \langle (u^{-1} \alpha_{i_p})^\vee, \lambda \rangle \geq 0$. It remains to exclude $c_p = 0$. If $c_p = 0$ then $s_{i_p} u\lambda = u\lambda$, hence $w\lambda = s_{i_1} \cdots s_{i_{p-1}} u\lambda$, and $s_{i_1} \cdots s_{i_{p-1}} u$ has length at most $\ell - 1$. This contradicts the minimality of $\ell(w)$ among elements sending λ to μ .

(ii) If $\langle \alpha_i^\vee, \mu \rangle \leq 0$ for every i then μ is antidominant, hence $\mu = w_0\lambda$. So assume $\langle \alpha_i^\vee, w\lambda \rangle > 0$. Then $\langle \lambda, w^{-1} \alpha_i^\vee \rangle > 0$; since λ is dominant this forces $w^{-1} \alpha_i > 0$, and therefore $\ell(s_i w) = \ell(w) + 1$. Let $J = \{j : \langle \alpha_j^\vee, \lambda \rangle = 0\}$, so that $W_\lambda = W_J$ and w is minimal in wW_J if and only if $w\alpha_j > 0$ for all $j \in J$. For $j \in J$ we have $w\alpha_j > 0$, and $s_i(w\alpha_j) < 0$ could only happen if $w\alpha_j = \alpha_i$; but then $\langle \alpha_i^\vee, w\lambda \rangle = \langle \alpha_j^\vee, \lambda \rangle = 0$, contradicting $\langle \alpha_i^\vee, w\lambda \rangle > 0$. Hence $s_i w\alpha_j > 0$ for all $j \in J$ and $s_i w$ is minimal in its coset. $\square$

Proof of Theorem 1.2. Fix a reduced word for w and extend it, by Lemma 4.1, to a reduced word for w_0^λ ; let σ_{full} be its c -weighted sequence and σ the prefix belonging to w . Let $X_0 = 0 \subseteq X_1 \subseteq \cdots \subseteq X_L = \operatorname{soc}_{\sigma_{\text{full}}}(I_\kappa)$ be the resulting filtration, indexed by the blocks $i_p^{c_p}$ of σ_{full} rather than by individual letters.

By Corollary 3.1 applied to the first letter of the p -th block, and because the remaining letters of the block can only enlarge X further,

$$\dim X_p - \dim X_{p-1} \geq c_p \quad (1 \leq p \leq L). \quad (2)$$

Here $c_p \geq 1$ by Lemma 4.1(i); it is this positivity that makes (2) a lower bound strong enough to be squeezed. Summing, and using that the c -trace telescopes, $\sum_p c_p \alpha_{i_p} = \lambda - w_0^\lambda \lambda = \lambda - w_0 \lambda$, we get

$$\dim \text{soc}_{\sigma_{\text{full}}}(I_\kappa) \geq \sum_p c_p = \dim(\lambda - w_0 \lambda) = \dim I_\kappa,$$

the last equality by Proposition 2.2. On the other hand $\text{soc}_{\sigma_{\text{full}}}(I_\kappa) \subseteq I_\kappa$, so the inequality is an equality. Consequently every inequality in (2) is an equality, and since the p -th block only enlarges X at vertex i_p ,

$$\underline{\dim} X_p - \underline{\dim} X_{p-1} = c_p \alpha_{i_p} \quad \text{for all } p.$$

Truncating at the prefix σ gives $\underline{\dim} \text{soc}_\sigma(I_\kappa) = \sum_{p \leq \ell} c_p \alpha_{i_p} = \lambda - w \lambda = v$. By Proposition 2.3 the submodule of I_κ with this dimension vector is unique and equals V . Word-independence follows, since the right-hand side depends only on w . $\square$

Corollary 4.2. *At the first letter of each block i^c of σ one has $\text{top}_i(X) = 0$ and $\dim \text{soc}_{(i)}(I_\kappa/X) = c$ exactly. (Within a block this fails, and must: after the first letter $\text{top}_i = c$ and $\text{soc}_{(i)}(I_\kappa/X) = 0$, as computed in Remark 3.2.)*

Remark 4.3. Corollary 4.2 has a purely combinatorial shadow. Only reflections at i and at its neighbours change $\langle \alpha_i^\vee, \cdot \rangle$, and a colour- i step flips that pairing from c to $-c$; so between two consecutive colour- i blocks of a c -trace some neighbouring colour must intervene. It is exactly that intervening step which absorbs S_i out of the head of X .

Proof of Proposition 1.5. If λ is minuscule then $\langle \alpha^\vee, \nu \rangle \in \{0, \pm 1\}$ for every root α and every weight ν of $L(\lambda)$, so every c_p equals 1 and w is λ -minuscule in the sense of [4, Def. 5.1]. By Stembridge [9, Prop. 2.1, §2] a λ -minuscule element is fully commutative, so its heap $H(w)$ is independent of the reduced word; and λ -minuscule heaps are ranked, with each colour fibre $H(w)_i$ a chain. By [5, Prop. 3.8, Lem. 3.10] the fixed point representation is the heap module $\mathbb{C}H(w)$, with V_i spanned by $H(w)_i$. The rank function of $H(w)$ is the path-length degree of Proposition 2.6, and a chain occupies pairwise distinct ranks, so every graded piece of V_i has dimension at most one. $\square$

5. READING THE COLOURED POSET

Let $X = \text{soc}_\sigma(I_\kappa) \cong V$ and assume the criterion of Corollary 1.3 holds, so that $\dim X_i[d] \leq 1$ for all i, d ; only then are the vertices of the poset canonically labelled. Define a coloured graded set

$$P(w) = \{(i, d, \text{basis vector of } X_i[d])\},$$

coloured by i and ranked by $d_{\max} - d$, with $(i, d) < (j, d+1)$ whenever j is adjacent to i and the corresponding arrow of Π is nonzero on the relevant vectors. Reading the degrees in decreasing order recovers the Hasse diagrams of [4, §5.4], whose top row is the top degree $d_{\max}$. Then [4, Thm. 5.6, Thm. 5.7] express the vertex function as a sum over reverse plane partitions on $P(w)$.

Example 5.1 ([4, Ex. 5.10]). $\Gamma = D_4$, $\lambda = \omega_2$, $w = s_1 s_2 s_3 s_4 s_2$. The c -trace gives $v = (2, 2, 1, 1)$ and $\sigma = (2, 4, 3, 2, 1, 1)$. The construction returns rows $[2], [1, 3, 4], [2], [1]$ read from the top degree down, matching the diagram on p. 82 of [4] including its six covering relations.

Example 5.2 ([4, Ex. 5.11, Ex. 5.12]). $\Gamma = D_5$, $\lambda = \omega_2$. For $w = s_2 s_1 s_3 s_4 s_5 s_3 s_2$ the construction returns $[2], [1, 3], [2, 4, 5], [3], [2]$, and for $w = s_2 s_1 s_2 s_3 s_4 s_5 s_3 s_2$ it returns $[2], [1, 3], [2, 4, 5], [3], [2], [1]$, both matching p. 83 of [4] with all covering relations.

6. TYPE D

Throughout this section $\Gamma = D_n$ with Bourbaki labelling $\alpha_i = e_i - e_{i+1}$ for $i \leq n-1$ and $\alpha_n = e_{n-1} + e_n$, so that the trivalent node is $t = n-2$.

Lemma 6.1. *Let $\lambda = \omega_k$ with $k \leq n-2$ and $\mu = w\lambda = \sum_{i \in S} \eta_i e_i$. With m and r as in Theorem 1.6 one has $v_t = r + 2m$.*

Proof. Write $\lambda - \mu = \sum_j c_j e_j$. Comparing coefficients of e_j in $\lambda - \mu = \sum_i v_i \alpha_i$ gives $c_1 = v_1$ and $c_j = v_j - v_{j-1}$ for $2 \leq j \leq n-2$, so $v_t = \sum_{j \leq n-2} c_j$. Since $k \leq n-2$ the support of λ lies in the first $n-2$ coordinates, whence $\sum_{j \leq n-2} c_j = k - \sum_{j \leq n-2} \mu_j$. Writing p and m for the number of positive and negative signs of μ among the first $n-2$ coordinates, $\sum_{j \leq n-2} \mu_j = p - m$ and $p + m = k - r$, so $v_t = k - (p - m) = r + 2m$. $\square$

Lemma 6.2 (the recursion). *Let $a_d(i) = \dim(I_\kappa)_i[d]$. Then for every i and every d strictly below the top degree,*

$$a_{d+1}(i) = \sum_{j \sim i} a_d(j) - a_{d-1}(i), \quad a_0(i) = \delta_{i, \kappa'}, \quad a_{-1} = 0,$$

where $\kappa' = \nu^{-1}(\kappa)$.

Proof. By construction the degree- $(d+1)$ part of the free module at j is $\bigoplus_{i \sim j} (I_\kappa)_i[d]$, and the preprojective relations impose one condition for each basis vector of $(I_\kappa)_j[d-1]$. These conditions are independent precisely when $Y : (I_\kappa)_j[d-1] \rightarrow \bigoplus_{i \sim j} (I_\kappa)_i[d]$ is injective. A nonzero element of $\ker Y$ would span a copy of S_j inside I_κ ; but I_κ has simple socle S_κ , concentrated in the top degree. Hence Y is injective below the top degree. $\square$

Lemma 6.3 (method of images). *For $\Gamma = D_n$ and $\lambda = \omega_k$ with $k \leq n-2$, the colour-t multiplicity sequence of I_κ is $(1, 2, \dots, 2, 1)$ with $k-1$ twos, occupying the $k+1$ degrees $t-k, t-k+2, \dots, t+k$. More generally vertex i carries multiplicity 2 in exactly $\max(0, i+k-n+1)$ degrees and multiplicity ≤ 2 throughout.*

Proof. The datum $a_0 = \delta_k$ is fixed by the diagram automorphism exchanging $n-1$ and n , hence so is the solution: $a_d(n-1) = a_d(n) =: b_d$. Setting $a_d(t+1) := 2b_d$, the recursion of Lemma 6.2 becomes the free chain recursion $a_{d+1}(i) + a_{d-1}(i) = a_d(i-1) + a_d(i+1)$ on $1, \dots, L$ with $L = t+1 = n-1$, subject to a Dirichlet condition $a_d(0) = 0$ and a Neumann (mirror) condition $a_d(L+1) = a_d(L-1)$.

On the infinite chain the solution with $u_0 = \delta_k, u_{-1} = 0$ is $u_d(i) = 1$ if $|i-k| \leq d$ and $i \equiv k+d \pmod{2}$, and 0 otherwise (induction on d). The Dirichlet condition is an odd reflection about 0 and the mirror condition an even reflection about L ; the group they generate is infinite dihedral, with images of k at $2Lj \pm k$ carrying alternating signs. Hence

$$a_d(i) = \sum_{x \text{ image of } k} \text{sign}(x) \cdot [|i-x| \leq d \text{ and } i \equiv x+d \pmod{2}].$$

The four images nearest t are k and $2L-k$ with sign $+$, at distances $t-k$ and $t+2-k$, and $-k$ and $2L+k$ with sign $-$, at distances $t+k$ and $t+2+k$. (The image solution vanishes identically once d exceeds $t+2+k$, and the total $\sum_{d,i} a_d(i)$ it predicts equals $\lambda - w_0 \lambda = \underline{\dim} I_\kappa$ by Proposition 2.2; this pins the top degree and shows that the recursion of Lemma 6.2, valid below the top degree, determines I_κ completely.) They enter the window one at a time as d grows, so $a_d(t)$ is 0 for $d < t-k$, then 1, then 2 on $[t+2-k, t+k)$, then 1 on $[t+k, t+2+k)$, then 0; the degrees of the correct parity in the middle range number $(2k-4)/2 + 1 = k-1$. The same computation at a vertex $i \leq t$ of the chain shows the value 2 occurs exactly for

$\max(|i - k|, 2L - k - i) \leq d < \min(i + k, 2L + k - i)$, giving $\max(0, i + k - n + 1)$ degrees. (At the two leaves $n - 1, n$ the folding gives $a_d(n - 1) = a_d(n) = b_d$ with $2b_d = a_d(t + 1) \leq 2$, so $b_d \leq 1$ and they never carry multiplicity 2; the displayed count applies only on the chain $i \leq t$.) $\square$

Remark 6.4 (window containment fails, and why). It is tempting to argue Theorem 1.4 in type D from Lemma 6.3 alone: the multiplicity-2 window of vertex $i \leq t$ is $[2L - k - i, i + k]$ with $L = n - 1$, whose endpoints satisfy $2L - k - i \geq L - k + 1$ and $i + k \leq (L - 1) + k$, so it is contained in the window of t *as an interval*. This is not enough, for two independent reasons.

First, the degrees carried by vertex i obey the parity constraint $d \equiv i + k \pmod{2}$, so the actual degree *sets* need not be nested when $i \not\equiv t \pmod{2}$. In D_6 with $\kappa = 3$ and $t = 4$, vertex 3 carries multiplicity 2 exactly at degree 4 while vertex t carries it exactly at degrees 3, 5; the sets are disjoint. (The same happens in D_7 , $k = 3$.) Second, even granting containment, knowing that a repeat at i occurs at a degree in the t -window does not produce a repeat at t : nothing transports $\dim V_i[d] \geq 2$ to $\dim V_t[d'] \geq 2$.

The mechanism that does work is inward injectivity, Lemma 7.1 below, which moves a repeat one vertex towards t and shifts the degree by one — and with that shift the D_6 numbers do line up, $i = 3$ at $d = 4$ mapping to t at $d = 5$. Lemma 7.1 is stated and used in §7; it is what proves Theorem 1.4 in both types.

Proposition 6.5 (top truncation). *Let $N \subseteq I_\kappa$ be a graded submodule with $\dim N = v = \lambda - w\lambda$ for some $w \in W$. Then the colour- t multiplicities of N are the greedy top truncation of the sequence of Lemma 6.3 to total v_t : the same degrees, filled from the highest downward.*

Note that the sequence $(1, 2^{k-1}, 1)$ of Lemma 6.3 is palindromic, so its top truncations and its bottom truncations have the same multiset of multiplicities, namely

$$v_t = 0 \mapsto (), \quad 1 \mapsto (1), \quad 2 \mapsto (1, 1), \quad 3 \mapsto (1, 2), \quad 4 \mapsto (1, 2, 1), \quad 5 \mapsto (1, 2, 2), \dots$$

Only the multiset is used below.

Proof for $k = 2$, checked for $n \leq 7$. Here the colour- t profile of I_κ is $(1, 2, 1)$ at degrees $d_0, d_0 + 2, d_0 + 4$ with $d_0 = n - 4$. Computing the submodule generated by a single colour- t element shows that its colour- t content is all of $(1, 2, 1)$ if the element sits in degree d_0 , is $(1, 1)$ in degrees $d_0 + 2, d_0 + 4$ if it sits in degree $d_0 + 2$ (for every choice of vector there), and is (1) if it sits in degree $d_0 + 4$. Hence the only colour- t profiles a submodule can have are $(0, 0, 0)$, $(0, 0, 1)$, $(0, 1, 1)$, $(0, 2, 1)$, $(1, 2, 1)$, and each is a top truncation. No hypothesis on v is needed. The generation pattern was computed for D_4 – D_7 and is uniform in n ; we have not written that induction out, but nothing depends on it, since Theorem 6.18 gives Proposition 6.5 for every k . $\square$

Remark 6.6 (how Proposition 6.5 is proved). Call a graded submodule $M \subseteq I_\kappa$ a *staircase* if its colour- t multiplicities are the greedy top truncation of those of I_κ — zero below one degree, proper at it, full above — and if at each of the two leaves M occupies a final segment of degrees. Everything left open in this paper is the following single statement.

Proposition 6.7 (the staircase property). *If $\dim M$ is w -extremal then M is a staircase. Equivalently, by the rigidity property of [2] quoted in §2, a submodule that is not a staircase satisfies $\delta(\dim M) > 0$, and so is never the unique submodule of its dimension vector.*

The proof occupies the rest of this section and is completed in Theorem 6.18: the statement is first made word-free, then reduced to k explicit weights, then to the closure of an explicitly written subspace, and finally to an inequality between two integer sequences.

Proposition 6.7 gives Proposition 6.5 at once: an extremal v carries a unique submodule, namely $V = \text{soc}_\sigma(I_\kappa)$ by Theorem 1.2, and the staircase property applies to it. It is verified on all 1124 graded submodules of I_κ over D_4 – D_7 : exactly 78 fail to be staircases and every one has $\delta = 2$ (`master.py`).

Proposition 6.8. *Proposition 6.7 holds whenever $v_{n-1} = v_n$.*

Proof. Suppose M is not a staircase at t , say $M_t[d] \neq 0$ while $M_t[d+2] \subsetneq (I_\kappa)_t[d+2]$. Every nonzero $x \in M_t[d]$ is then *special* — annihilated by an arrow from t to a leaf — since otherwise the three double paths $a_{j \rightarrow t} a_{t \rightarrow j} x$, which satisfy one linear relation and hence span at most two dimensions, would by direct computation span all of $(I_\kappa)_t[d+2]$, forcing it into M . The special vectors form two lines rather than a subspace, so $M_t[d]$ is one of them, and $\dim(I_\kappa)_t[d] = 2$ at every such degree. The diagram automorphism τ exchanging the leaves fixes κ (as $k \leq n-2$), hence acts on I_κ , and exchanges the two arrows out of t to the leaves, hence the two special lines; so $\tau(M) \neq M$. When $v_{n-1} = v_n$ we have $\dim \tau(M) = \dim M$, and two distinct submodules of one dimension vector contradict rigidity, so $\dim M$ is not extremal and $\delta > 0$ by Corollary 2.4. $\square$

The statement now reduces to a single assertion at a single vertex. Carry along the construction the hypothesis

$$(H) \quad X \text{ is an up-set at } t \text{ and at the two leaves,}$$

where *up-set* means zero below one degree, proper at it, full above; at t this is top-truncation, and at a leaf, where $\dim(I_\kappa)_j[d] \leq 1$, it says X_j occupies a final segment. Steps at other colours do not change X there, so (H) need only be checked at colour- t and leaf steps.

Leaf steps. Let j be a leaf, whose only neighbour is t . Then $c_j = v'_t - 2v'_j \geq 1$ because the word is reduced. The leaf degrees are $g_1 + 1 > \dots > g_k + 1$ with $g_i = t + k - 2i$, so under (H) the largest unfilled one is $D_j = t + k - 2v'_j - 1$, while $D = t + k - 2a$ if $v'_t = 2a$ and $D = t + k - 2a - 2$ if $v'_t = 2a + 1$. In the first case $c_j \geq 1$ forces $v'_j \leq a - 1$, in the second $v'_j \leq a$; either way $D_j > D$. As X is an up-set at t it is full above D , hence at $D_j + 1$, so the socle condition $a_{j \rightarrow t} \xi \in X_j[D_j + 1]$ is automatic and the top unfilled leaf slot lies in $\text{soc}_{(j)}(I_\kappa/X)$ — as does every unfilled leaf slot at a degree $\geq D$, a consecutive block at the top. So (H) is preserved at leaf steps.

Colour- t steps. Preservation at t is (G1) and (G2), and both follow from (H) , the bound $c \geq 1$, and two inequalities about the profile of I_κ alone:

$$\sum_{d \geq D+1} \dim(I_\kappa)_{t-1}[d] \leq 2a - [k = t] \quad \text{and} \quad \leq 2a + 2 - [k = t]$$

in the two cases. These are statements about the profile of Lemma 6.3 alone, and the strand form of it proves them by counting. At $t-1$ the $+$ strand runs over $[|t-1-k|, t+k-3]$ and the $-$ strand over $[t-k+3, 2t-|t-1-k|]$, in steps of 2, each with $\min(t-1, k)$ slots. In the first case $D+1 = t+k-2a+1$, which has the parity of $t-1+k$, so the $+$ strand contributes $\max(0, a-1)$ slots above it and the $-$ strand $a+1-[k=t]$, capped at $\min(t-1, k)$; in the second $D+1 = t+k-2a-1$ and the contributions are a and $a+2-[k=t]$. Summing gives $2a - [k=t]$ and $2a + 2 - [k=t]$, with equality in the generic range and the cap only helping. (Checked at all 164160 instances over D_5 – D_{80} and every k .) At the leaves, $c \geq 1$ gives $\mu'_{n-1} \leq 0$, and then $v'_{n-1} + v'_n = v'_t - \mu'_{n-1}$ with $|v'_{n-1} - v'_n| = |\mu'_n| \leq 1$ yields $\min(v'_{n-1}, v'_n) \geq a$ in the (G1) case, and in the (G2) case exactly one leaf deficient by one dimension when $r' = 0$ and none when $r' = 2$.

What is left. Both (G1) and (G2) also need information about the chain neighbour $t - 1$ at one degree, and that is all that remains:

- (*) at a colour- t step, if $X_t[D] \neq 0$ then $X_{t-1}[D + 1]$ is full,
and if $X_t[D] = 0$ it is deficient by at most one dimension.

This is strictly weaker than an up-set at $t - 1$: X_{t-1} is full in *all* degrees $> D$ at only 2207 of 2337 colour- t steps, whereas (*) holds at all of them, so degree $D + 1$ is exactly the right place to look.

Half of (*) is proved. Suppose $X_t[D] \neq 0$, so it is spanned by a special vector x , say $a_{t \rightarrow n}x = 0$. Then $a_{t \rightarrow (t-1)}x \neq 0$: otherwise x is killed by two of the three arrows out of t , and not by the third, since I_κ has simple socle S_κ and $\kappa \leq n - 2$ is not a leaf; so $y = a_{t \rightarrow (n-1)}x \neq 0$, and the relation at t ,

$$\epsilon_1 a_{(t-1) \rightarrow t} a_{t \rightarrow (t-1)}x + \epsilon_2 a_{(n-1) \rightarrow t}y + \epsilon_3 a_{n \rightarrow t} a_{t \rightarrow n}x = 0,$$

has vanishing first and third terms, giving $a_{(n-1) \rightarrow t}y = 0$ — a nonzero element of the leaf $n - 1$ killed by its only arrow, hence in $\text{soc}(I_\kappa) = S_\kappa$, forcing κ to be that leaf. So $a_{t \rightarrow (t-1)}(X_t[D])$ is a nonzero line in $X_{t-1}[D + 1]$, and (*) holds whenever $\dim(I_\kappa)_{t-1}[D + 1] = 1$ — which is 422 of the 547 instances with $X_t[D] \neq 0$.

Splitting the 2337 colour- t steps by branch and by $r' = |\text{supp } \mu' \cap \{n - 1, n\}|$ gives a clean table:

branch	n	full	deficient by 1	by more
$X_t[D] \neq 0$	547	547	0	0
$X_t[D] = 0, r' = 0$	855	855	0	0
$X_t[D] = 0, r' = 1$	935	805	130	0

The first row is what (G1) requires and the other two what (G2) requires: the deficiency never exceeds one and occurs only in the $r' = 1$ branch, which is precisely where both leaves are full, so at most one neighbour is ever deficient. What remains is therefore (A) the first row when $\dim(I_\kappa)_{t-1}[D + 1] = 2$, namely 125 of the 547; (B) the second row; and (C) the third row, which is automatic unless that dimension is 2.

All three concern the single space $X_{t-1}[D + 1]$, and each would follow from the up-set property at $t - 1$, which is the statement

- (*)' at every colour- t socle step, X is an up-set at $t - 1$,

verified at 2179/2179 steps (`upsets.py`). What makes (*)' tractable is that the construction order can be removed from it entirely. At such a step $X = V(w')$ for a prefix w' of the reduced word, so X is itself an extremal module by Theorem 1.2; and the step having colour t means $c_t = \langle \alpha_t^\vee, \mu' \rangle \geq 1$ for $\mu' = \lambda - \underline{\dim} X$. Hence (*)' follows from

- (*)^o every extremal $M \subseteq I_\kappa$ with $\langle \alpha_t^\vee, \lambda - \underline{\dim} M \rangle \geq 1$
is an up-set at $t - 1$,

which mentions neither w nor any order of construction. The side condition is exactly right: over D_5 – D_{10} and all $\kappa \leq n - 2$ in range, (*)^o holds at 1998/1998 extremal modules, while among those failing to be up-sets at $t - 1$ — there are 111 — *every* one has $c_t = 0$ (`starchar.py`).

(*)^o can now be reduced to finitely many weights, uniformly in the rank. Let $J = I \setminus \{t - 1, t\}$ and let $j \in J$ be a descent of μ . Put $\nu = s_j \mu$, which is shorter. By Theorem 1.2, $V(\mu) = \text{soc}_j V(\nu)$, and a socle step of colour j enlarges the module only at vertex j ; as $j \neq t - 1$,

$$V(\mu) \text{ and } V(\nu) \text{ have the same graded profile at } t - 1.$$

So the property “up-set at $t - 1$ ” is constant on W_J -orbits, and $(*)^\circ$ for μ follows from $(*)^\circ$ for the W_J -minimal representative of its orbit. (One cannot take $j = t$: there $\langle \alpha_t^\vee, s_t \mu \rangle = -\langle \alpha_t^\vee, \mu \rangle \leq -1$, so $s_t \mu$ is longer.)

Proposition 6.9. *The W_J -minimal $\mu \in W\lambda$ with $\langle \alpha_t^\vee, \mu \rangle \geq 1$ are exactly those whose only descent is $t - 1$, namely, in Bourbaki coordinates,*

$$\mu_a = e_1 + \cdots + e_a - (e_{n-2-c} + \cdots + e_{n-3}) + e_{n-2}, \quad a + c = k - 1, \quad 0 \leq a \leq k - 1.$$

There are k of them, independently of n — or $k - 1$ when $k = n - 2$, where $a = k - 1$ gives $\mu = \lambda$.

Proof. For $k \leq n - 2$, $W\omega_k = \{\sum_{i \in S} \eta_i e_i : |S| = k, \eta_i = \pm 1\}$, so $\mu = (\eta_1, \dots, \eta_n)$ with $\eta_i \in \{0, \pm 1\}$ and exactly k of them nonzero, and $\langle \alpha_j^\vee, \mu \rangle = \eta_j - \eta_{j+1}$ for $j \leq n - 1$, $\langle \alpha_n^\vee, \mu \rangle = \eta_{n-1} + \eta_n$. The hypotheses read $\eta_j \geq \eta_{j+1}$ for every $j \leq n - 1$ with $j \neq n - 3$, together with $\eta_{n-1} + \eta_n \geq 0$ and $\eta_{n-2} - \eta_{n-1} \geq 1$. If $\eta_{n-1} = -1$ then $\eta_n \leq -1$, contradicting $\eta_{n-1} + \eta_n \geq 0$; if $\eta_{n-1} = 1$ then $\eta_{n-2} \geq 2$, impossible. So $\eta_{n-1} = 0$, whence $\eta_n \leq 0$ and $\eta_n \geq 0$, so $\eta_n = 0$ and $\eta_{n-2} = 1$. On $1, \dots, n - 3$ the sequence is weakly decreasing, hence of the form $1^a 0^b (-1)^c$ with $a + b + c = n - 3$, and counting nonzeros gives $a + c + 1 = k$. Conversely each such μ_a has its unique descent at $n - 3 = t - 1$. Finally $b = n - 2 - k \geq 0$, so a runs over $0, \dots, k - 1$. $\square$

The reduction is verified on the modules themselves in `reduce.py`: 15364 instances of $(*)^\circ$ over D_4 – D_{10} , $k \leq 6$, collapse to 83 representatives, the profile at $t - 1$ agreeing across every reduction step (1705/1705).

At the representatives more is true than $(*)^\circ$ asks: $V(\mu_a)$ is the greedy top truncation of I_κ at *every* vertex, not only at t where Proposition 6.5 puts it. So what remains is

- (B) for the k weights μ_a of Proposition 6.9, the graded multiplicities of $V(\mu_a)$ are the greedy top truncation of those of I_κ , at every vertex,

which gives $(*)^\circ$ at once, a greedy top truncation being an up-set. (B) holds at every representative in computable range (43/43). Its point is that it is a bounded list: the same assertion for arbitrary extremal μ is false — see below — and even for arbitrary leaf-balanced extremal μ with $c_t \geq 1$ it fails, at 62 of 348 instances over D_4 – D_8 . The earlier route through a chainwide staircase invariant (`staircase.py`) required in addition control of how colour- $(t - j)$ steps interleave with colour- t steps along the c -trace; nothing of the kind is needed here.

(B) can be turned into a closure check on an explicitly written subspace. A graded top truncation is not determined by its multiplicities: where I_κ carries multiplicity 2 and the truncation takes only 1, a line must be chosen. Over all extremal modules exactly three lines ever occur in such a slot, and the representatives always use the same one, canonically described: τ , the diagram automorphism exchanging the leaves, fixes I_κ (as κ is a chain vertex), acts on every multiplicity-2 slot with eigenvalues ± 1 , and the line used is the $+1$ eigenline; the other two are the special lines, the kernels of the two leaf arrows, which τ exchanges. Define $T(v)$, the *canonical top truncation*, to be full above the threshold, zero below, and the τ -eigenline at a partially taken multiplicity-2 slot. Then $\dim T(v) = v$ by construction, so by rigidity

$$(B) \iff T(v_a) \text{ is a submodule of } I_\kappa,$$

and the socle construction disappears from the statement altogether.

Closure is then almost formal, given three facts about I_κ and τ alone — no words, no dimension vectors:

- (T0) every arrow between graded slots has rank $\min(\dim \text{source}, \dim \text{target})$: it is injective or surjective, whichever is possible (450/450);
- (T1) a rank-1 arrow into a multiplicity-2 chain slot has image the τ -eigenline (22/22);
- (T2) the τ -eigenline of a multiplicity-2 chain slot is annihilated by exactly the arrows into 1-dimensional *chain* slots — zero there (22/22), nonzero into 2-dimensional chain slots (36/36) and into both leaves (44/44).

Every closure obligation for $T(v_a)$ is then of a resolved kind: full into full (nothing to check); full into partial (the arrow has rank 1 by (T0) and its image is the eigenline by (T1)); eigenline into full; eigenline into partial (the image is τ -fixed, hence in the target eigenline, by equivariance); and eigenline into an empty slot (the image vanishes by (T2)). Whether an obligation is of one of these kinds depends only on the two slot dimensions, on whether the target is a leaf, and on the greedy thresholds — so all of (B) beyond (T0)–(T2) is arithmetic.

These three have a common source, and with it they stop being conjectural. At a chain vertex τ splits every graded piece into eigenspaces — write $a_d^\pm(j)$ for the eigen-multiplicities — and that splitting is as clean as it could be. With $t = n - 2$, so that the top degree is $2t$:

- (W1) each eigenspace is multiplicity-free in every chain slot, so a multiplicity-2 slot is exactly one + line and one – line (8255/8255);
- (W2) in closed form,

$$a_d^+(j) = 1 \iff |j - k| \leq d \leq j + k - 2 \text{ and } d \equiv j + k \pmod{2}, \quad a_d^-(j) = a_{2t-d}^+(j),$$

the + strand being the Dirichlet half-line wave of the generator and the – strand its reflection in the top degree. The two add up to the multiplicity of Lemma 6.3 (27430/27430) and are the eigen-multiplicities (8255/8255);

- (W3) hence each strand occupies a contiguous run of degrees at every chain vertex, the + strand the low ones and the – strand the high ones, overlapping exactly in the multiplicity-2 window (2236/2236);
- (W4) every arrow restricts to an *isomorphism* between consecutive nonzero slots of the same strand (17706/17706).

So I_κ is two multiplicity-free waves travelling through the chain, the arrows transporting each (`strands.py`).

Of these, (W1) and (W3) are immediate from the closed form (W2), and (W2) itself is a consequence of Lemma 6.2.

Proposition 6.10. *(W2) holds; hence so do (W1) and (W3).*

Proof. The complex of Lemma 6.2 at a chain vertex j , $(I_\kappa)_j[d-1] \rightarrow \bigoplus_{i \sim j} (I_\kappa)_i[d] \rightarrow (I_\kappa)_j[d+1]$, consists of τ -equivariant maps: the arrows between chain vertices commute with τ , and for $j = t$ the two leaf summands are exchanged by it, so their sum is τ -stable. As $\text{char} \neq 2$ the complex splits into eigen-parts, and exactness and the injectivity of the first map below the top degree are inherited by each summand. Writing b_d for the common dimension of $(I_\kappa)_{n-1}[d]$ and $(I_\kappa)_n[d]$, the pair is the induced representation of $\mathbb{Z}/2$, so each eigen-part of it has dimension b_d . Counting dimensions in the ϵ -part therefore gives, for every chain vertex j and $0 < d < 2t$,

$$a_{d+1}^\epsilon(j) + a_{d-1}^\epsilon(j) = \sum_{i \sim j, i \leq t} a_d^\epsilon(i) + [j = t] b_d,$$

while Lemma 6.2 at a leaf reads $b_{d+1} + b_{d-1} = a_d^+(t) + a_d^-(t)$. The generator e_k spans $(I_\kappa)_k[0]$ and is fixed by τ , so $a_0^+(j) = \delta_{jk}$, $a_0^- \equiv 0$ and $b_0 = 0$. Each equation expresses degree $d+1$ in terms of degrees d and $d-1$, so this system has at most one solution, determined by induction on d ; and the closed form of (W2), with $b_d = a_d^+(t+1)$, satisfies it. (That last verification

is arithmetic in n, k, j, d , and holds over D_5 – D_{60} for every k at all 11 443 152 instances of the chain recursion and 130 032 of the leaf recursion, with no exception.) $\square$

(W4) is a consequence of the same τ -refined Lemma 6.2, through a propagation along the chain. Write $[\text{lo}_j^\epsilon, \text{hi}_j^\epsilon]$ for the interval (W3) attaches to the strand ϵ at the chain vertex j , so that $\text{hi}_j^+ = j + k - 2$ and $\text{hi}_j^- = 2t - |j - k|$. The single arithmetic fact used is

$$\min(\text{hi}_{j-1}^\epsilon, \text{hi}_{j+1}^\epsilon) = \text{hi}_j^\epsilon - 1, \quad (3)$$

immediate for $+$, and for $-$ because $\max(|j - 1 - k|, |j + 1 - k|) = |j - k| + 1$.

Lemma 6.11 (a death propagates). *Let $1 < j < t$ and let x span the ϵ -line at (d, j) . Suppose $a_{j \rightarrow i}x = 0$ for one chain neighbour i of j , the slot $(d + 1, i)$ carrying ϵ . Write i' for the other chain neighbour. Then $a_{j \rightarrow i'}x$ spans the ϵ -line at $(d + 1, i')$, the slot $(d + 2, j)$ carries ϵ , and $a_{i' \rightarrow j}(a_{j \rightarrow i'}x) = 0$.*

Proof. As $j < t$ its only neighbours are i and i' , so the ϵ -part of Lemma 6.2 forces $a_{j \rightarrow i'}x \neq 0$; it lies in the ϵ -part of $(I_\kappa)_{i'}[d + 1]$, which is at most a line by (W1), so that slot carries ϵ too. Both slots $(d + 1, j \pm 1)$ therefore carry ϵ , whence $d + 1 \leq \min(\text{hi}_{j-1}^\epsilon, \text{hi}_{j+1}^\epsilon) = \text{hi}_j^\epsilon - 1$ by (3), i.e. $d + 2 \leq \text{hi}_j^\epsilon$; and $d + 2 \geq d \geq \text{lo}_j^\epsilon$ with the parity unchanged, so $(d + 2, j)$ carries ϵ . Finally the relation at j reads $\epsilon_1 a_{i \rightarrow j} a_{j \rightarrow i}x + \epsilon_2 a_{i' \rightarrow j} a_{j \rightarrow i'}x = 0$, whose first term vanishes. $\square$

Proposition 6.12 (outward arrows). *For either strand, $a_{j \rightarrow (j+1)}$ is nonzero on the ϵ -line at (d, j) whenever $(d + 1, j + 1)$ carries ϵ .*

Proof. Suppose $a_{j \rightarrow (j+1)}x = 0$. If $j = 1$ the only chain neighbour is 2 and the ϵ -part of Lemma 6.2 is contradicted at once. Otherwise Lemma 6.11 with $i = j + 1$, $i' = j - 1$ reproduces the situation at $(d + 1, j - 1)$: its outward arrow dies and the target $(d + 2, j)$ carries ϵ . Iterating lowers the vertex to 1. $\square$

Proposition 6.13 (the folded leaf). *The map $J[d]^\epsilon \rightarrow (I_\kappa)_t[d + 1]^\epsilon$, $J = (I_\kappa)_{n-1} \oplus (I_\kappa)_n$, is nonzero whenever both sides are.*

Proof. The leaves are multiplicity-free, so $(I_\kappa)_{n-1}[d]$ is a line, spanned by u say, and J^ϵ is spanned by $(u, \epsilon\tau u)$, whose image is $(1 + \epsilon\tau)z$ with $z = a_{(n-1) \rightarrow t}u$. As $n - 1$ has the single neighbour t , Lemma 6.2 makes $z \neq 0$, and the relation at $n - 1$ reads $a_{t \rightarrow (n-1)}z = 0$. If $\dim(I_\kappa)_t[d + 1] = 1$ the claim is exactly $z \neq 0$.

So let that slot be 2-dimensional and suppose z were a τ -eigenvector, of eigenvalue ϵ . Applying τ to $a_{t \rightarrow (n-1)}z = 0$ gives $a_{t \rightarrow n}z = 0$, so both leaf arrows kill z ; by Lemma 6.2 at t , $y := a_{t \rightarrow (t-1)}z \neq 0$, and y lies in the ϵ -part at $(d + 2, t - 1)$, which is therefore nonzero. The relation at t applied to z has both leaf terms zero, so $a_{(t-1) \rightarrow t}y = 0$. Finally $(d + 3, t)$ carries ϵ : for $\epsilon = +$ because $d + 2 \leq \text{hi}_{t-1}^+ = \text{hi}_t^+ - 1$; for $\epsilon = -$ because the slot $(d + 1, t)$ carries $+$ as well, so $d + 1 \leq \text{hi}_t^+ = t + k - 2$ and hence $d + 3 \leq t + k = \text{hi}_t^-$. Thus the outward arrow at $(d + 2, t - 1)$ dies with its target carrying ϵ , contradicting Proposition 6.12. So z is not an eigenvector and both of its eigen-components are nonzero. $\square$

Corollary 6.14. *(W4) holds.*

Proof. Outward arrows are Proposition 6.12. For an inward arrow, a death at (d, j) towards $j - 1$ produces by Lemma 6.11 a death at $(d + 1, j + 1)$ towards j , again inward, and iterating raises the vertex to t . There the relation carries the two leaf terms: the death makes the composite through the leaves vanish on the ϵ -line, while Lemma 6.2 makes its image in J^ϵ nonzero — contradicting Proposition 6.13. $\square$

With (W1)–(W4) in place, (T0), (T1) and (T2) are unconditional.

Proposition 6.15. *(W) implies (T0), (T1) and (T2) for arrows between chain vertices.*

Proof. By (W4) the rank of an arrow $(d, j) \rightarrow (d+1, i)$ is the number of strands carried by both slots, so (T0) says exactly that the two sets of strands present are *nested*. Suppose not; then one slot carries only $+$ and the other only $-$. Throughout $i, j, k \leq t$ and $|i - j| = 1$.

Case A: $+$ only at (d, j) , $-$ only at $(d+1, i)$. Since $+$ holds at (d, j) we have $d \geq |j - k| \geq |i - k| - 1$, so $d+1 \geq |i - k|$ and the failure of $+$ at $(d+1, i)$ must be $d+1 > i+k-2$, i.e. $d \geq i+k-2$; while $+$ at (d, j) gives $d \leq j+k-2$. If $i = j+1$ this reads $d \leq i+k-3 < i+k-2 \leq d$. If $i = j-1$ then $d \in \{i+k-2, i+k-1\}$ and the parity $d \equiv j+k = i+k+1$ forces $d = i+k-1 = j+k-2$; now $-$ at $(d+1, i) = (i+k, i)$ needs $i+k \geq 2t-i-k+2$, i.e. $i+k \geq t+1$, while the absence of $-$ at (d, j) needs $j+k-2 < 2t-j-k+2$, i.e. $j+k \leq t+1$ — impossible for $j = i+1$.

Case B: $-$ only at (d, j) , $+$ only at $(d+1, i)$. From $+$ at $(d+1, i)$, $d \leq i+k-3$. The absence of $+$ at (d, j) is either $d > j+k-2$, which with $d \leq i+k-3$ forces $j < i-1$, or $d < |j-k|$; in the latter case $-$ at (d, j) gives $2t-j-k+2 \leq d < |j-k|$, which reads $2t+2 < 2j$ if $j \geq k$ and $2t+2 < 2k$ if $j < k$, both impossible.

For (T1) the source carries one strand and the target two; suppose the source strand were $-$. From $+$ at $(d+1, i)$, $d \leq i+k-3$. The absence of $+$ at (d, j) is $d > j+k-2$, contradicted by $d \leq i+k-3 \leq j+k-2$, or $d < |j-k|$, excluded exactly as in Case B.

For (T2) the source carries both strands and the target one; suppose the target strand were $+$. The absence of $-$ at $(d+1, i)$ is either $d \leq 2t-i-k$, which against $d \geq 2t-j-k+2$ from $-$ at (d, j) gives $i \leq j-2$; or $d \geq 2t-|i-k|$, which against $d \leq j+k-2$ from $+$ at (d, j) gives $2t+2 \leq i+j$ when $i \geq k$ and $2t+2 \leq j-i+2k \leq 2t+1$ when $i < k$. All are impossible. $\square$

The nesting statement, and (T1) and (T2) with it, are checked directly from the closed form over D_5 – D_{100} , every k , every adjacent pair and every degree: 23 517 840 nesting instances and 156 848 each of (T1) and (T2), with no exception (`strands.py`).

In that form everything can be tested far past the range where the modules can be built, and the residue reduces to two integer sequences. With (W) proved, a τ -invariant graded submodule splits at chain vertices as $N^+ \oplus N^-$ with N^ϵ a subobject of the strand ϵ ; and since (d, j) reaches $(d+2, j)$ by an outward step followed by an inward one, a subobject meets each vertex in a *top interval*. So such a submodule is determined by the numbers $\dim N_j^\epsilon$, and closure of $T(v_a)$ says exactly that its two strand-parts are up-closed.

Each strand has $M_j = \min(j, k)$ slots at a chain vertex j , of which $o_j = \max(0, j+k-t-1)$ are shared with the other (26486/26486). Greedy from the top fills the $M_j - o_j$ slots carrying only the $-$ strand, then the o_j shared ones two dimensions at a time — an odd one taking the τ -line, which is the $+$ strand — and then the $+$ -only slots, so the occupation numbers are

$$(s_j^+, s_j^-) = \begin{cases} (0, v_j), & v_j \leq M_j - o_j, \\ (\lceil r/2 \rceil, M_j - o_j + \lfloor r/2 \rfloor), & r := v_j - (M_j - o_j) \leq 2o_j, \\ (v_j - M_j, M_j), & \text{otherwise,} \end{cases}$$

which reproduces the truncation exactly (459282/459282). Writing $\text{hi}_j^+ = j+k-2$ and $\text{hi}_j^- = 2t-|j-k|$ for the top of each strand, the lowest occupied degree is $\text{hi}_j^\epsilon - 2(s_j^\epsilon - 1)$, so up-closedness at an edge $j \rightarrow j+1$ is precisely

$$0 \leq s_{j+1}^+ - s_j^+ \leq 1, \quad |s_{j+1}^- - s_j^-| \leq 1.$$

Write $u_j = v_j - \min(j, k)$, so that $s_j^+ = \max(0, u_j, \lceil (u_j + o_j)/2 \rceil)$ (459282/459282) and $s_j^- = v_j - s_j^+$. Almost all of the two inequalities now falls out.

Proposition 6.16. $0 \leq s_{j+1}^+ - s_j^+ \leq 1$ for every chain edge; and $|s_{j+1}^- - s_j^-| \leq 1$ except possibly where v jumps by 2.

Proof. Reading off $x_p = \lambda_p - (\mu_a)_p$ region by region gives $u_{j+1} - u_j \in \{-1, 0, 1\}$ always, and $= -1$ only for $j+1 \leq a$ or $j+1 = t$; also $o_{j+1} - o_j \in \{0, 1\}$. Each of the three terms in s^+ increases by at most one under such a step, so $s_{j+1}^+ - s_j^+ \leq 1$. For the lower bound, only the two exceptional steps can lower a term. For $j+1 \leq a$ one has $v_j = 0$ and $u_j + o_j \leq k - t - 1 < 0$, so s^+ is 0 at both ends. For $j+1 = t$, o steps up while u steps down, so $u_j + o_j$ is unchanged and only the middle term u falls; but $\lceil (u + o)/2 \rceil < u$ forces $u \geq o + 2$, whereas $u_{t-1} \leq o_{t-1} + 1$ (34104/34104), so the maximum is not attained by u alone and s^+ does not drop. Finally $s_{j+1}^- - s_j^- = x_{j+1} - (s_{j+1}^+ - s_j^+)$ with $x_{j+1} \in \{-1, 0, 1, 2\}$; the values 0 and 1 are harmless, and $x_{j+1} = -1$ occurs only at $j+1 = t$, where s^+ is unchanged by the above. $\square$

The one implication left over is proved by the same bookkeeping.

Lemma 6.17. If $x_{j+1} = 2$ then $s_{j+1}^+ - s_j^+ = 1$.

Proof. $x_p = 2$ exactly for $a + b < p \leq \min(k, t - 1)$, where $b = t - k$. So $j = p - 1$ satisfies $a + b \leq j \leq \min(k, t - 1) - 1$; in particular $j < k$, so $\min(j, k) = j$, $\min(j + 1, k) = j + 1$ and $u_{j+1} - u_j = x_{j+1} - 1 = 1$. In this range $v_j = b + 2(j - a - b)$, hence $u_j = j - 2a - b$ and $o_j = \max(0, j - b - 1)$.

If $j \geq b + 1$ then $o_j = j - b - 1$ and $o_{j+1} = j - b$, so $u + o$ increases by 2 and $\lceil (u + o)/2 \rceil$ by exactly 1. Moreover $u_j - o_j = 1 - 2a \leq 1$, so $u_j \leq o_j + 1$ and the middle term does not strictly beat $\lceil (u_j + o_j)/2 \rceil$; and $u_j + o_j = 2(j - a - b) - 1 \geq -1$, so that term is ≥ 0 . Hence $s_j^+ = \lceil (u_j + o_j)/2 \rceil$ and $s_{j+1}^+ \geq s_j^+ + 1$, which with Proposition 6.16 is equality.

If $j \leq b$ then $a + b \leq j \leq b$ forces $a = 0$ and $j = b$, so $u_j = 0$ and $o_j = o_{j+1} = 0$, giving $s_j^+ = 0$ and $u_{j+1} = 1$, whence $s_{j+1}^+ = 1$. $\square$

The obligations that involve a leaf are settled by three identities at the node. At t one has $v_t = 2k - 2a - 2$ and each leaf carries $v = k - a - 1$ (34104/34104); $\min(t, k) = k$ and $o_t = k - 1$, so the truncation at t takes $k - a$ slots and its lowest taken degree, which is also its threshold D_t , equals $t - k + 2a + 2$ (32508/32508); while the leaves, being multiplicity-free with top degree $t + k - 1$, start at $t - k + 2a + 3$ (32508/32508) — exactly one higher. So every taken slot at t maps into a taken leaf slot, and every taken leaf slot maps into the slot of degree $D_t + 2$ at t , which is full. Both leaf patterns are therefore “target full”, and require nothing (1462902/1462902 instances).

Theorem 6.18. Proposition 6.7 holds, and with it Theorem 1.6.

Proof. By Proposition 6.9 it suffices to prove (B) at the k representatives, and by rigidity that is the assertion that the canonical truncation $T(v_a)$ is a submodule. Its closure obligations along the chain are the up-closedness of its two strand-parts, which by Proposition 6.16 and Lemma 6.17 is the pair of increment bounds; those involving a leaf land in full slots by the identities above. (T0), (T1) and (T2), which identify the images, are Proposition 6.15. $\square$

Every step above is re-checked in `audit.py`, which walks the chain from the main theorem to the closed form and prints one line per link. The counts, over D_5 – D_{45} with every k and every a unless stated otherwise: $\dim \text{soc}_\sigma(I_\kappa) = v$ and agreement with the independent solver on 277 elements; Lemma 7.1 at 315 triples in type E (the complete finite verification) and 55

in type D ; Lemma 6.3 at 880 profiles; $v_t = r + 2m$ at 24312 elements; (W1)–(W4) at 3570, 3570, 1140, 7300 slots and the folded-leaf arrow at 635; (T0), (T1), (T2) at 2 922 500, 32508, 32508 instances over D_5 – D_{60} ; the representatives at 24 pairs (n, k) ; $T(v_a)$ a submodule and equal to $V(\mu_a)$ at 32; the arithmetic at 8 047 275 up-closure instances, 445 178 increments of each kind, 148 994 drops, 84273 jumps of two and 446 982 leaf obligations, with the parity case occurring 0 times; (G1) and (G2) at 164 160 instances over D_5 – D_{80} ; and the criterion itself at 5819 elements over D_4 – D_{10} . The audit also runs negative controls — statements that must *fail*, including the one-neighbour interval step that an earlier draft of Lemma 6.11 wrongly used — checks that the criterion is unchanged at $p = 5, 7, 11, 13$ and 1000003, and that it does not depend on the reduced word.

Extremality cannot be dropped from $(*)^\circ$. The candidate localisation — for every graded $N \subseteq I_\kappa$, (H) and $c_t \geq 1$ imply N is an up-set at $t-1$ — holds for every graded submodule with $\kappa \leq 3$ and fails at $\kappa = 4$, where $t-1$ first carries multiplicity 2 in two different degrees. The smallest counterexample is D_6 , $\kappa = 4$, $t = 4$, with graded profile $(\dim N / \dim I_\kappa \text{ by degree})$

$$\begin{array}{ll} 1 : 0/1 & 1/1 & 4 : 0/1 & 0/2 & 1/2 & 2/2 & 1/1 \\ 2 : 0/1 & 1/2 & 1/1 & 5 : 0/1 & 1/1 & 1/1 & 1/1 \\ 3 : 0/1 & 1/2 & 1/2 & 1/1 & 6 : 0/1 & 0/1 & 1/1 & 1/1 \end{array}$$

which satisfies (H) , has $c_t = 1$, and is partial at two degrees at $t-1$ (`localstar.py`).

It is worth recording what does *not* work, since it is the first thing one tries. The hypothesis that V is simply the greedy top truncation of I_κ at *every* vertex — which would be a closed formula and would settle everything at once — is false. It holds at 1185 of 1307 extremal modules over D_4 – D_7 , E_6 , E_7 and A_5 , the smallest failure being D_5 , $\lambda = \omega_2$, $\mu = (1, -1, 0, 0, 0)$, where V_2 occupies degrees 2 and 6 while the top truncation would give 6 and 4; vertex 2 is the chain neighbour $t-1$. That element has $r + 2m = 2$ and so does have simple spectrum, so the failure is not confined to degenerate cases. Top truncation at t and at the two leaves — the hypothesis (H) — does hold for every extremal module tested.

Two facts delimit the argument. It genuinely needs extremality: the staircase property fails for 78 of the 1124 graded submodules above, and the analogous statement for arbitrary submodules is false. And the obvious mechanism is not available: when $X_t[D]$ is a line it is *always* one of the two special lines (527/527), so one leaf arrow annihilates it and that leaf’s fullness at $D+1$ comes from elsewhere in X .

Proof of Theorem 1.6. For the spin weights λ is minuscule and Proposition 1.5 applies. Let $k \leq n-2$. By Lemma 6.3 the colour- t profile of I_κ is $(1, 2^{k-1}, 1)$. By Theorem 1.2, V is a submodule of I_κ with $\dim V = v$, so Proposition 6.5 applies to it and the colour- t multiplicities of V are the truncation displayed above. Every multiplicity is ≤ 1 exactly when $v_t \leq 2$. By Theorem 1.4, simplicity at t implies simplicity everywhere. Lemma 6.1 identifies $v_t = r + 2m$. $\square$

Remark 6.19. The disjunctive form “ $c_{\max} = 1$ or $v_t \leq 2$ ” of the criterion, natural from the c -trace, collapses once the spin weights are separated: for non-spin λ in type D , $c_{\max} = 1$ forces $(r, m) \in \{(0, 0), (0, 1), (1, 0), (2, 0)\}$, all of which satisfy $r + 2m \leq 2$.

7. TYPE E

Theorem 1.2 and Corollary 1.3 are uniform and therefore settle type E as well: the criterion is decidable from any reduced word. Two further points deserve comment.

7.1. Reduction to the trivalent node. As Remark 6.4 explains, no containment-of-windows argument proves Theorem 1.4, in either type: in E_6 with $\kappa = 3$ (so $I_3 = \Pi e_5$) vertex 3 carries

multiplicity 2 in path-length degree 4 while vertex 4 is zero there, and the same phenomenon already occurs in D_6 with $\kappa = 3$. The mechanism that does work is the following.

For $i \neq t$ let $\text{nx}(i)$ denote the neighbour of i one step closer to t .

Lemma 7.1 (inward injectivity). *Let Γ be E_6 , E_7 or E_8 . For every fundamental weight κ , every $i \neq t$ and every degree d with $\dim(I_\kappa)_i[d] \geq 2$, the arrow*

$$(I_\kappa)_i[d] \longrightarrow (I_\kappa)_{\text{nx}(i)}[d+1]$$

is injective.

Proof. The three diagrams have finitely many triples (κ, i, d) in total, so this is a finite verification: 11 instances in E_6 , 55 in E_7 and 249 in E_8 , all injective. See `inwardlemma.py`. $\square$

Proof of Theorem 1.4 in type E. Suppose $\dim V_i[d] \geq 2$ for some $i \neq t$. Then $\dim(I_\kappa)_i[d] \geq \dim V_i[d] \geq 2$, so by Lemma 7.1 the inward arrow is injective on $(I_\kappa)_i[d]$, hence on the subspace $V_i[d]$. Since V is a submodule of I_κ , the image lies in $V_{\text{nx}(i)}[d+1]$, so $\dim V_{\text{nx}(i)}[d+1] \geq 2$. Iterating $\text{dist}(i, t)$ times gives $\dim V_t[d + \text{dist}(i, t)] \geq 2$. Contrapositively, if V_t is multiplicity-free then so is V . $\square$

In type D the same statement holds, and there it can be proved outright, uniformly in the rank.

Lemma 7.2 (Lemma 7.1 in type D). *Let $\Gamma = D_n$, $t = n - 2$, let $i < t$ be a chain vertex and let d satisfy $\dim(I_\kappa)_i[d] \geq 2$. Then the inward arrow $(I_\kappa)_i[d] \rightarrow (I_\kappa)_{i+1}[d+1]$ is injective.*

Proof. Suppose $x \neq 0$ lies in the kernel. For $i < t$ the vertex i has exactly the two neighbours $i-1$ and $i+1$. By Lemma 6.2 the socle of I_κ is simple and concentrated in the top degree, so an element killed by every arrow out of i is either zero or that socle generator; since $\dim(I_\kappa)_i[d] \geq 2$ we are below the top degree, and therefore

$$y := a_{i \rightarrow i-1}(x) \neq 0.$$

The preprojective relation at i reads $\varepsilon a_{i+1 \rightarrow i} a_{i \rightarrow i+1} + \varepsilon' a_{i-1 \rightarrow i} a_{i \rightarrow i-1} = 0$ on $(I_\kappa)_i[d]$; applied to x , whose first term vanishes by hypothesis, it gives $a_{i-1 \rightarrow i}(y) = 0$. So y is a nonzero element of $(I_\kappa)_{i-1}[d+1]$ killed by its inward arrow: the same situation, one vertex further from t and one degree higher.

Iterating $i-1$ times produces a nonzero $z \in (I_\kappa)_1[d+i-1]$ killed by the arrow $1 \rightarrow 2$. In D_n the vertex 1 has 2 as its only neighbour, so z is killed by every arrow out of 1 and hence lies in $\text{soc}(I_\kappa) = S_\kappa$, forcing $\kappa = 1$. But by Lemma 6.3 a chain vertex i carries multiplicity 2 in exactly $\max(0, i+k-n+1)$ degrees, which vanishes for $k=1$ and $i \leq n-2$; so $\dim(I_\kappa)_i[d] \geq 2$ is impossible when $\kappa = 1$. Contradiction. $\square$

Remark 7.3. The two leaves $n-1, n$ need no argument: by the folding in Lemma 6.3 they satisfy $a_d(n-1) = a_d(n) = b_d$ with $2b_d = a_d(t+1) \leq 2$, so $b_d \leq 1$ and Lemma 7.2 is vacuous there. With Lemma 7.2 in hand the proof of Theorem 1.4 given above applies verbatim in type D , for every rank. It is also confirmed directly for D_4 – D_{10} (126 instances) by `inwardlemma.py`, and the two inputs — simple socle and the relation at a chain vertex — by `inwardproof.py`.

7.2. No closed form of type- D shape. What does *not* carry over is the closed form of Theorem 1.6: v_t is no longer a complete invariant. In E_6 with $\lambda = \omega_3$ both simple and non-simple elements occur at $v_t = 3$; likewise in E_7 with $\lambda = \omega_2$ at $v_t = 4$. A closed form must therefore involve more than the multiplicity at the trivalent node. We have not found one.

The colour- t multiplicity profiles of I_κ in type E are finite data, computed once and for all; in E_6 they are $(1, 1, 1, 1)$, $(1, 1, 2, 1, 1)$, $(1, 2, 2, 2, 1)$, $(1, 2, 3, 3, 2, 1)$, $(1, 2, 2, 2, 1)$, $(1, 1, 1, 1)$ for

$k = 1, \dots, 6$, and the maximum multiplicity anywhere in I_κ is 6, attained in E_8 at $k = 4$. This is also why the truncation mechanism of §6 is specific to type D : it turns on $\dim(I_\kappa)_t[d] \leq 2$, so that the three double paths at t , subject to one relation, can span everything.

8. VERIFICATION

The accompanying code reproduces every claim; each script runs standalone and prints a pass/fail line.

- **proof.py** — the five steps of §4. The Crawley-Boevey identity holds at 5517/5517 socle steps; $\dim I_\kappa = \lambda - w_0\lambda$ at 47/47 injectives across $A_3, A_4, A_5, D_4, D_5, D_6, D_7, E_6, E_7$; $\dim \text{soc}_\sigma(I_\kappa) = v$ for 706/706 elements; the minuscule statement for 256/256; Proposition 1.4 for 981/981 elements, of which 681 are of type E and the rest of type D .
- **socdim.py** — Theorem 1.2 on 1149 elements and 12829 individual socle steps, no exceptions.
- **sweepall.py** — the aggregate criterion-versus-solver sweep quoted above; DKKCAP raises the orbit-size cutoff and DKKP changes the characteristic.
- **verify.py** — the printed posets of [4, pp. 82–83] reproduced with their Hasse edges; the criterion against an independent graded-submodule solver; word-independence; Theorem 1.6.
- **truncatest.py** — Proposition 6.5 (top truncation) on 1000 elements of D_4 – D_7 , every non-spin fundamental weight ($k \leq n - 2$) with orbit at most 400, no exceptions.
- **inwardlemma.py** — Lemma 7.1, exhaustively in E_6, E_7, E_8 (315 instances) and for D_4 – D_{10} (126 instances).
- **inwardproof.py** — the inputs to Lemma 7.2: simple socle in the top degree (39 injectives, D_4 – D_9), multiplicity-freeness on the chain for $\kappa = 1$, and the relation at a chain vertex (550 instances).
- **bigrade.py** — Lemma 2.5: Γ is a tree and the potential f exists, for A_2 – A_8, D_4 – D_9, E_6, E_7, E_8 .
- **master.py** — Proposition 6.7 over all 1124 graded submodules of I_κ for D_4 – D_7 : exactly 78 non-staircases, every one with $\delta = 2$.
- **starchar.py** — $(*)^\circ$ at 1998/1998 extremal modules over D_5 – D_{10} , with all 111 up-set failures at $t - 1$ having $c_t = 0$.
- **localstar.py** — the failure of the localisation of $(*)^\circ$: 947/951 of the graded submodules of I_κ satisfying its hypotheses over D_4 – D_7 , the four exceptions all at $\kappa = 4$.
- **reduce.py** — Proposition 6.9 and (B): 15364 instances over D_4 – D_{10} , $k \leq 6$, collapsing to 83 representatives; reduction step 1705/1705, (B) at 43/43.
- **canon.py** — τ , the canonical truncation $T(v)$, (T0)–(T2), and the classification of the closure obligations.
- **strands.py** — the strand decomposition (W1)–(W4), the closed form, and the arithmetic steps of Lemma 6.11 and Proposition 6.13, over D_5 – D_{17} and every κ .
- **audit.py** — the whole chain re-checked link by link, with negative controls, characteristic and word-independence, and a check that the counts quoted here are the ones it computes.
- **thresh.py** — the closed forms for the profile of I_κ and for v_a , checked against the modules, and the obligation classification over D_5 – D_{60} : 34104 representatives, 26.2 million obligations and 2.99 million instances of (T0)–(T2), all resolved.
- **filltop.py, claimG.py, epscrit.py, violators.py** — the unconditional reductions (a), (b), (c) of Remark 6.6 and the structure of the non-staircases.

- `squeeze.py`, `homological.py`, `headrule.py`, `lemA.py`, `eprojan.py` — the individual ingredients.

Aggregate agreement of Corollary 1.3 with the independent solver is 2179/2179, broken down as type A 214, type D 1232, type E 733; this is the output of `sweepall.py`, which enumerates every fundamental weight of A_3 – A_6 , D_4 – D_7 , E_6 , $E_7\omega_1$ and $E_7\omega_7$ whose Weyl orbit has at most 400 elements and compares the criterion against the solver element by element. All computations are over a finite prime field; the whole suite was rerun at $p = 5, 11, 13$ with identical results.

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